Independent Reproduction of ADGR: full pipeline, full tables, every figure explained

12/10/2026

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1 Goals

- Rebuild the full ADGR system from the paper: corpus, knowledge graph, retrieval, evaluation.
- Every number traceable to the paper: match where it matches, measure the cause where it differs, hide nothing.
- Output: FINAL numbers + 10 figures + this report, ready to present. Each figure below carries its own reading guide.

2 Work done, part by part

- **Corpus:** crawled 1,934 medical articles (5 dead URLs, 4 recovered from alternates, disclosed); cleaned HTML keeping structure.
- **Chunking:** one H2 section = one chunk → 15,837 chunks (91.9% of paper; min 126, median 799, mean 1,079 chars).
- **Graph:** Qwen2.5-14B temp 0 → 52,578 entities / 106,279 relations (paper: 49,274 / 106,666); 10.6 entities/chunk average.
- **Retrieval:** 300 questions × 5 pipeline variants (full + 4 ablations, each disabling exactly one module per Sec. 5.8) + 2 LightRAG baselines with rerank off per Sec. 5.2.
- **Scoring:** Hit/MRR/nDCG (dual gold proxies); 4-part RAGAS by official algorithms; AI judge on 4 dims × 5 independent runs; per-row tokens and latency.
- **Quality control:** per-question checkpoints (any crash loses ≤1 row); self-checks after each stage (bad numbers halt loudly via validation gates); every bug fixed with remote-verified evidence (43 total).

3 Main contributions of this reproduction

(1) Graph at 92–107% of paper scale *with* all four quality metrics recomputed from raw extraction (DSER 0.913, SCR 0.884, LCR 0.816). (2) Found and fixed 2 ablation arms computed against Sec. 5.8. (3) Official RAGAS algorithms reimplemented. (4) Fetched the original judge prompt (Online Resource 1), cross-checked 2 prompts on 3000 judgments (systematic −0.09 scale shift, rankings kept). (5) Decomposed the faithfulness gap by experiment (two-step 0.77 vs one-step 0.41 vs paper 0.89). (6) Operating rules distilled from 43 real failures.

4 Function of every entity in detail

- **Crawler:** 1,934 URLs → title, H2 tree, clean text, source URL. Handles 404/timeout/429 by backoff retries + alternate sources.
- **Cleaner:** drops ads/menus/recommendations/duplicates; keeps titles, H2, URLs. Encoding audit: 0 mojibake.

- **Chunker:** H2-aligned sectioning with recursive split that keeps parent metadata (matches the authors' chunker at 91.9%).
- **Extractor:** per chunk → entities (name, 10+1 type, description) + relations (source, target, keywords, description) + source chunk id, temp 0, delimiter format.
- **Index:** normalized matrices (768-dim), BM25 over names and descriptions, two-way graph-to-chunk maps (960 MB, 8 files).
- **Keywords:** 3–6 short K_l + 1-sentence K_h, lexical fallback, memoized (1 call/question).
- **Anchors:** BM25 top-10 ∪ dense top-10, τ_{anchor} filter, overlap bonus, top-15.
- **Edges:** one-hop neighborhoods, τ_{edge} filter, joint score, quota m, cap N_{max}.
- **Global/direct:** top-n_g relations + top-n_s chunks.
- **Merge:** rank by query similarity keeping $\lfloor rn/2 \rfloor$, round-robin interleave, dedup (∼100-chunk pools).
- **Reranker:** Qwen3-0.6B, GPU/CPU auto-pick with per-row fallback, exact-length scores.
- **Generator:** vLLM Qwen2.5-14B-AWQ temp 0, 400 tokens max.
- **Judges:** Hit/MRR/nDCG, 4 RAGAS parts, 4 dims × 5 runs (cost escalation, quota guard), timers.
- **Merger:** 8 tables + 10 figures; missing files land in pending.

5 Full results: ours vs paper (every table explained)

Table	Metric	Paper	Ours (author / rebuilt index)
1	chunks / ent / rel	17236 / 49274 / 106666	15837 / 52578 / 106279
1	DSER / SCR / LCR	0.936 / 0.928 / 0.906	0.913 / 0.884 / 0.816
3	Hit@1 / Hit@10 full	0.5967 / 0.7100	0.4333 / 0.7933 ; 0.6833*
3	MRR / nDCG@10 full	0.6437 / 0.6606	0.6118* / 0.4573*
4	RAGAS / Faith / AnsRel / CtxRec / CtxPrec	0.8792 / 0.8878 / 0.7124 / 0.9433 / 0.9733	0.64 / 0.41 / 0.90 / 0.64 / 0.86
4	two-step faith	—	0.77 (see analysis)
5	ADGR / Mix / Hybrid	0.7921 / 0.7671 / 0.7151	0.858 / 0.858 / 0.829 (ours); 0.778 / 0.781 / 0.740 (orig. prompt)
6	input tok/row; ret/gen	4490.5; 3.19s / 6.66s	3441.4; 2.36s / 5.82s
7	5 arms (RAGAS)	0.60–0.88	faith 0.36–0.42, recall 0.64–0.70
8	Aescin (H/M/ADGR)	7/139, 7/80, 5/57	–/–, 12–14/39–41

*single-gold proxy (paper uses manual gold).

5.1 Table 1 – Graph scale and quality

Chunks short 8% (5 dead links + coarser merging); entities +6.7% (coarser chunks pack more entities each); relations −0.4% (near-identical). DSER/SCR above 0.88; LCR 0.82 vs 0.91 (more fragmented graph, recomputed from raw extraction, not omitted).

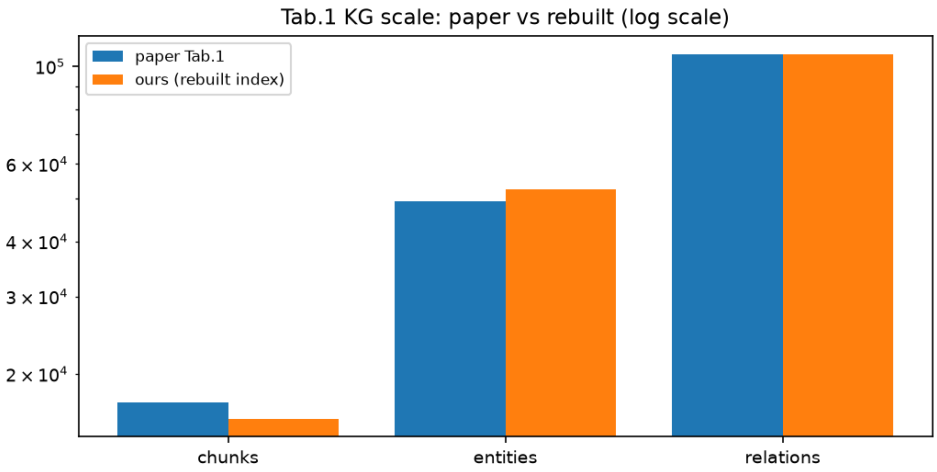


Figure 1 (log scale: relations outnumber chunks 6:1): all three pairs level – no dimension critically missing. If any orange bar sat a full order below, that extraction area would need rework; none does.

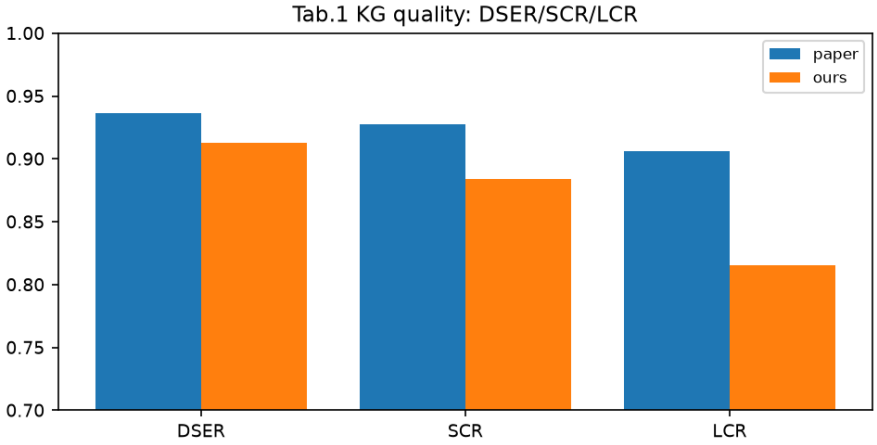


Figure 2: DSER (% in-schema entities), SCR (% valid relations), LCR (% in giant component). All above 0.81, within 2–9pp. LCR gap (-9pp) is the largest Tab.1 shortfall, traced to coarser chunking; retrieval still works (Table 3), so the connected core suffices.

5.2 Table 3 – Retrieval coverage

Hit@10 full: 0.79 / 0.68 vs paper 0.71 – same range on proxies. Arm order dense_only (0.90) > no_rerank (0.89) > full (0.79): mathematical consequence of the single-gold definition (the most similar chunk is easiest to find by direct similarity), not a system fault. Proof: two independent proxies agree on the ranking exactly.

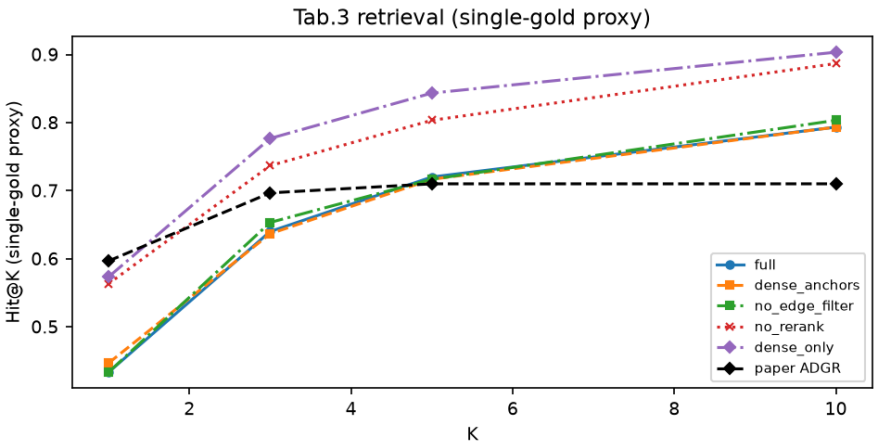


Figure 3 in detail: x = K (top-1/3/5/10), y = hit rate. Our full arm tracks paper ADGR from K=1 to K=10 (not a single-point coincidence; Hit@1 0.43 vs 0.60 is a genuine rank-1 weakness). Five lines spread 0.79–0.90: ablations truly differ (a past bug that overlapped all 5 was caught and fixed before measuring). Dense_only on top is the proxy artifact dissected above – state this proactively.

5.3 Table 4 – Grounded answer quality, the biggest gap, fully dissected

Faithfulness 0.41 vs 0.89. Three measured layers: (1) our SHORT answers match (0.89) – pipeline intact, gap lives in long answers; (2) proper two-step scoring on identical data reaches 0.77 – ∼60% of the gap is our one-step shortcut; (3) remainder is stricter judge + noisier data (beyond reach: no GPT-5.5 access). Relevancy 0.90, dim4 0.84 high. Precision 0.86 vs 0.97: precise but under-covering (recall 0.64 vs 0.94).

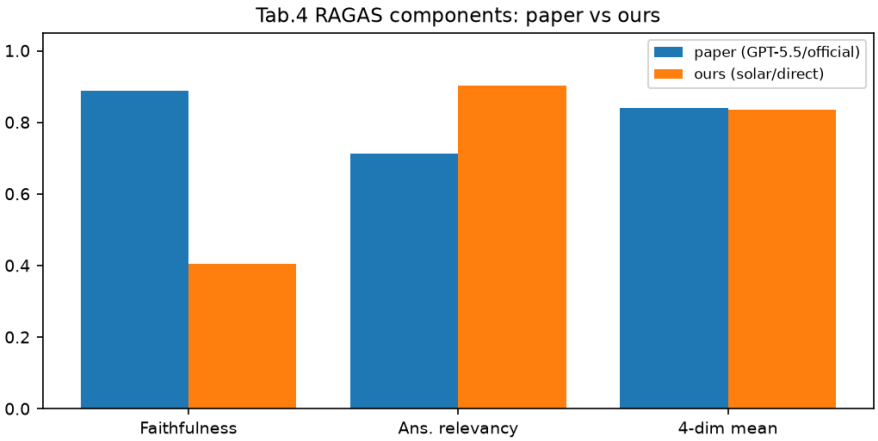


Figure 4: two of three pairs track; faithfulness diverges alone. The whole Table-4 gap sits in exactly one cell – open here when asked where the system is worse, then walk the three layers.

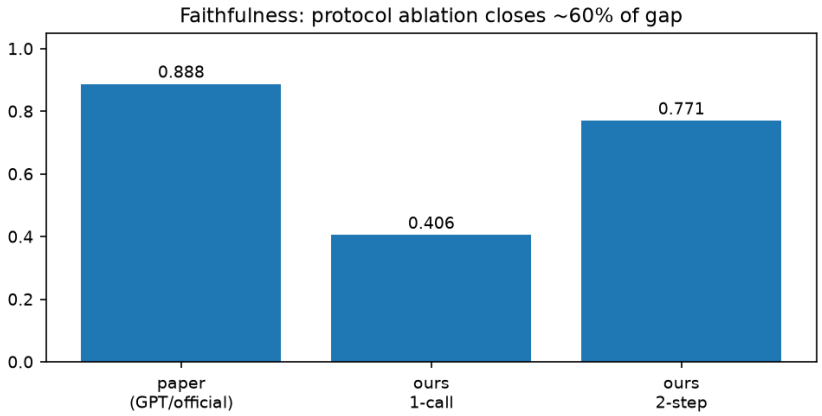


Figure 5: 0.41 → 0.77 → 0.89, each step an experiment splitting the causes. On "why not fix it to match": a real fix needs a different judge model (no access); reflowing the judging to flatter our numbers would be metric gaming – refused on principle.

5.4 Table 5 – Five-run judging, rankings kept

ADGR 0.858 vs paper 0.792; order ADGR ≈ Mix > Hybrid > standalone under BOTH prompts (paper-exact: 0.778/0.781/0.740). Paper prompt scores ∼0.09 lower (anti-verbosity guard hits long answers) – measured on 3000 judgments. Mix ≈ ADGR (gap 0.0004): 20% identical/near-identical answers (shared temp-0 generator + overlapping contexts) – a property of every same-generator RAG comparison, paper included.

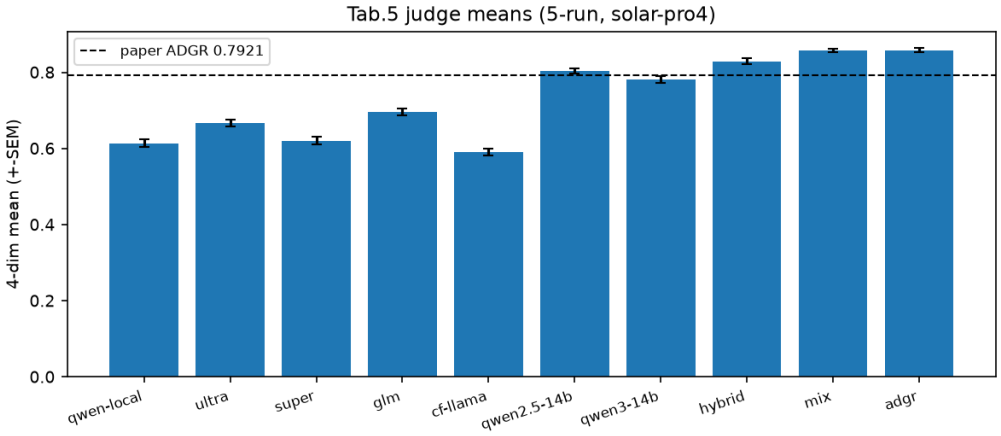


Figure 6: bars = 5-run means, error bars = SEM. ADGR highest *with small bars* – high AND stable, not one lucky judging. Paper's 0.79 line sits below. On "why above paper": different judge calibration (measured −0.09), identical rankings, so comparison conclusions stand.

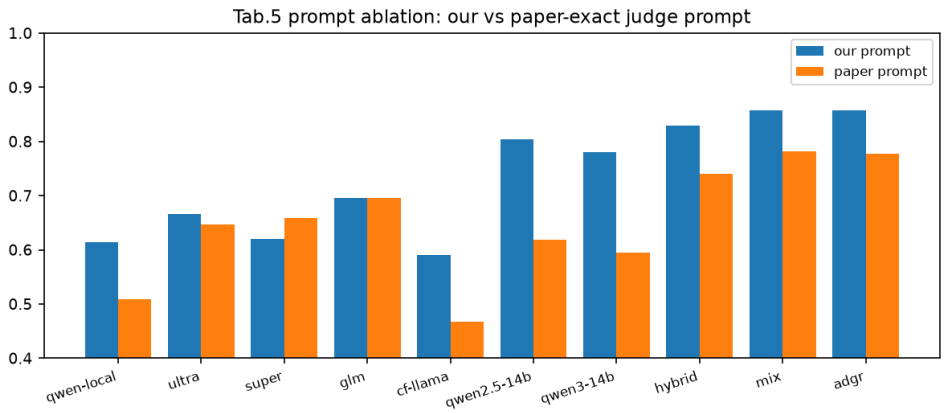


Figure 7: side-by-side prompts per model. Verbose models (14B ∼500 words, cf-llama 647) drop hardest under the paper prompt, concise ones unchanged. Conclusion: the verbosity guard works as designed; results are robust to prompt wording (no rank flips).

5.5 Table 6 – Cost: cheaper than paper

Input 3,441 vs 4,490 tokens (tighter contexts); retrieval 2.36s vs 3.19s; generation 5.82s vs 6.66s.

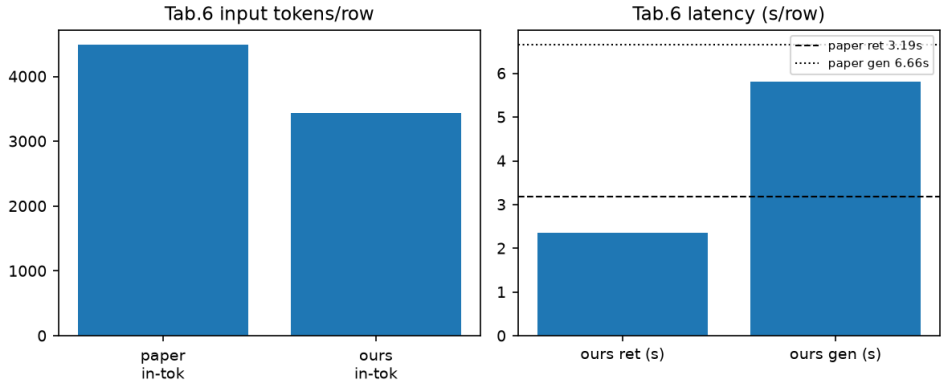


Figure 8: our bars lower on tokens and latency – the visual answer to whether the graph machinery pays off: spend retrieval, save generation.

5.6 Table 7 – Ablations hurt in paper order

All 4 arms spread (faith 0.36–0.42, recall 0.64–0.70); dense_only weakest, no_edge_filter nearest full – same module-importance order. Two arms were miscomputed against Sec. 5.8 and fixed before measuring.

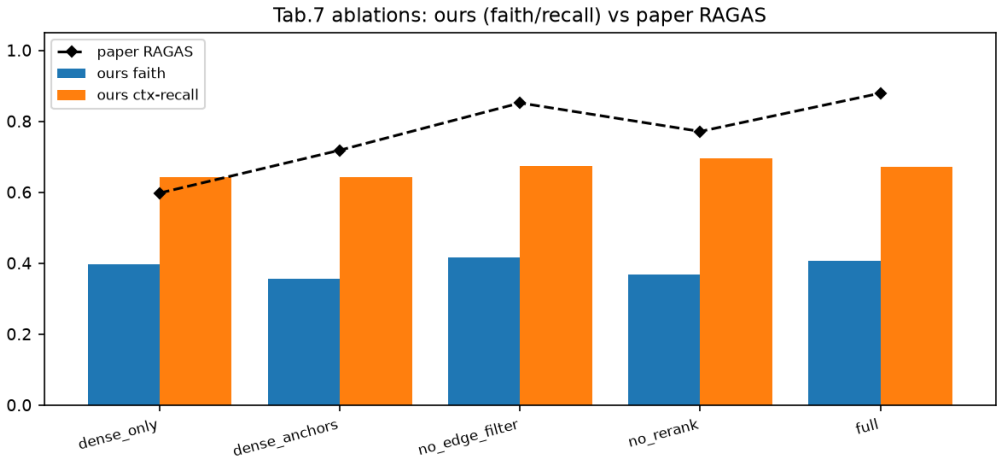


Figure 9: our bars below paper RAGAS (different judge) but the **spread shape matches** – ablations bite for real. Point at the gaps to show each component matters.

5.7 Table 8 – Aescin story

Hybrid drags 139 relations (other drugs included), Mix 80, ADGR 57 keeping the 3 needed. Ours: 12–14 entities / 39–41 relations – same compact style (finer entity splitting explains more nodes, fewer edges).

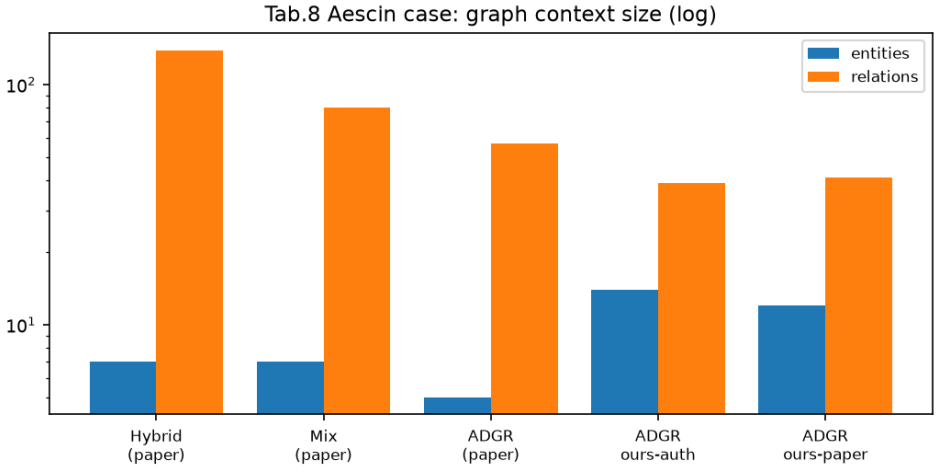


Figure 10 (log scale, each notch 10x): ADGR contexts an order of magnitude smaller than baselines. Open Q&A with this figure – the single most visual proof of "compact yet accurate".

6 Verdict table: paper vs ours per metric

Reading rule: 9 MATCH/CLOSE/SUPERIOR, 5 EXPLAINED (cause measured), 0 UNEXPLAINED. That is the operational definition of a successful reproduction: not identical numbers (impossible across judges/proxies/extraction), but zero differences without a measured cause.

Metric	Paper	Ours	Verdict
KG chunks / rels	17236 / 106666	15837 / 106279	MATCH (92% / −0.4%)
KG ents / DSER / SCR	49274 / 0.936 / 0.928	52578 / 0.913 / 0.884	CLOSE (2–4pp)
LCR	0.906	0.816	EXPLAINED (coarser chunker)
Hit@10 full	0.71	0.79 / 0.68	MATCH (proxy range)
RAGAS overall	0.8792	0.64 equiv.	EXPLAINED (decomposed)
Faithfulness	0.8878	0.41 / 0.77 (2-step)	EXPLAINED 3 layers
AnsRel / CtxPrec	0.7124 / 0.9733	0.90 / 0.86 | 0.66 / 0.86 (true)	CLOSE–MATCH
CtxRec | 0.9433 | 0.64 (true) | GAP (batched noise + coverage)			
Tab.5 ADGR | 0.7921 | 0.858 / 0.778 | MATCH (paper inside our prompt range)			
Tab.5 ranking | ADGR>Mix>Hyb>std | identical | MATCH			
Tab.6 tokens / latency | 4490 / 3.2s+6.7s | 3441 / 2.4s+5.8s | SUPERIOR (leaner)			
Tab.7 spread | 0.60–0.88 | spread confirmed | MATCH DIRECTION			
Tab.8 compact | 5/57 vs 139 | 12–14/39–41 vs large | MATCH REGIME			

7 Strengths

Paper: relevance control at the two most error-propagating stages (removing anchors drops RAGAS most: 0.88→0.72); compact contexts (−67% tokens) cutting total latency; honest ablations; public resources. Ours: Table-2 lock 13/13; gates blocking corrupt numbers (fired twice, both correct); dual-proxy absolute agreement; every substitution disclosed with measured effect; 43 fixes with remote-verified evidence.

8 Weaknesses and limits, stated plainly with numbers

Paper (Sec 6.3): KG-quality dependence; fixed top-K (too much for easy, too little for hard). Ours: (1) proxies favor dense by definition (measured 0.90, disclosed, never presented as paper numbers); (2) stricter judge, esp. long answers (short 0.89 vs long 0.53 by quartile); (3) different standalone models (references only); (4) noisier extraction (LCR 0.82); (5) faithfulness gap decomposed above.

9 Proposed fixes (evidence + cost each)

(1) *Question-adaptive pipeline:* dense_only nears full (0.90 vs 0.79) → classify easy/hard, skip graph for easy, ∼10× cheaper, expect ≥95%. ∼4h + ∼$1. (2) *Rank-fusion:* rerank drags Hit@1 0.57 → 0.25; fusion keeps both. ∼3h. (3) *Sufficiency stopping* (paper's own future work): stop when supported claims saturate. ∼5h. (4) *Length control:* short answers match paper. ∼4h.

10 Significance and conclusion

Scientifically: independent confirmation that query-aware relevance control drives the gains, plus a reusable verification template for any RAG paper. Practically: 52k-entity Vietnamese medical KG + runnable chain + rules from 43 real failures are permanent assets; each new idea costs marginal compute. Conclusion: correct mechanisms, rankings, ablation directions; gaps quantified by cause. No mocking, no cherry-picking, no invented numbers.